



THRUM

DRONE INSTRUMENT

Set a key, a chord and a temperament. That harmony maps onto the Launchpad X, and you build a drone out of it by hand — tones swell in over seconds, hold indefinitely, and ride up and down under your fingers.

Made to be improvised over.



What this thing is

A drone is not a chord you play, it is a chord you *inhabit*. Thrum holds one indefinitely so somebody else can explore it.

Thrum has thirty-two independent voices, one per pad of the Launchpad's upper grid. A voice is not a note: it has no decay, and letting go of one is a slow fade rather than a note-off. You tap tones in, they bloom over several seconds, and they stay until you take them away.

The eight columns of the grid are the **mode** — seven scale degrees plus the octave — and the four rows are four octaves of it. Every pad is in the scale, so nothing you press is wrong. That is the whole point: it should be safe to build a drone without looking.

SIXTY SECONDS TO A DRONE

- 1 Open Thrum. It starts in **D Dorian** — a Dm7 — in 5-limit just intonation, on the Harmonium timbre.
- 2 Click **Tanpura** under Voicings. Four tones swell in over about seven seconds.
- 3 Let it settle for half a minute. It should never sound quite the same twice.
- 4 Add or remove tones from the grid. ⌘-click one to give it a sitar buzz.
- 5 Under Pulse, press **Run**. The chord you are holding starts to arpeggiate through itself, two lanes at different rates. **TAP** four times to set the tempo by hand.
- 6 Or skip all of that: press the 🎵 in the top bar and Thrum plays itself. Different every time.
- 7 When you're done, ⏏ lets everything go gracefully.

WHY IT DOESN'T GET BORING

Every partial of every voice is **two oscillators** offset by a fraction of a hertz. The offsets are in hertz rather than cents on purpose: a 0.3 Hz detune beats three times every ten seconds whether it sits on the fundamental or the eleventh harmonic, so the whole spectrum breathes at one slow rate instead of the top of it going ragged.

Every voice's pitch drift, tremolo, filter sweep, pan and sitar swirl run off LFOs whose periods are distinct **prime numbers of seconds** — 11, 13, 17, 19, 23 and up. A voice with a 41-second drift and a 67-second filter sweep returns to the same state every 2,747 seconds. Two such voices, effectively never.

One thing worth knowing. Thrum's reverb can hold a thirty-second tail. After you silence everything with ⏏, the room keeps ringing for a long time — that is correct, not a stuck note. **PANIC** is the one control that also clears the tail.

WHAT YOU NEED

Nothing at all	The whole instrument is playable from the Mac window. The grid on screen mirrors the hardware exactly.
Novation Launchpad X	Recommended. Plug it in and Thrum takes it over automatically — no setup, no template to load. All 81 keys are used.
Novation Launch Control	Optional. Its knobs bind to Thrum's sliders on plug-in. Until then every control lives in the app.



Top bar

Chord readout	The current key and chord, e.g. Dm7 , coloured by mode. Underneath: the mode name, the temperament, and the reference pitch.
🕒 Flow	Hands the instrument to itself — see <i>Flow</i> . Turns slowly while it is running, in the current mode's colour.
DSP	Share of the audio deadline the engine is using, as a decaying peak. Under 10% is normal. If it shows drops in red, blocks are being missed and you will hear crackle. Click to reset the tally.
Output meter	Eighteen segments, peak-reading. The top three turn red.
Device pills	Green dot when a Launchpad X or Launch Control is connected. Hot-plugging either is fine.

KEY

Twelve pitch classes. The tonal centre is always placed at concert pitch; every interval above it comes from the temperament you choose.

Octave – 3 +	Register of the root, 1–6. The grid's lowest row sits an octave <i>below</i> this, so the default of 3 puts the bottom row at D2 and the top at D5.
A440 / A432	Reference pitch. A432 lowers everything by about 32 cents; some people find it settles better under a long drone. Sounding tones glide, they don't jump.



Mode · Chord

Sixteen modes, each paired with the chord it implies. Picking "Dorian" and picking "m7" are the same act — the chord is what the drone holds, the mode is what the soloist gets to explore over it. Chord tones are lit brighter on the grid than the rest of the scale.

Changing mode mid-drone is a modulation, not an edit. Sounding tones glide to their new pitches over half a second rather than jumping. Switching Dorian → Locrian under a held chord is a legitimate performance move.

BANK A — DIATONIC CORE	CHORD	DEGREES
Ionian	maj7	1 9 3 11 5 13 7
Dorian	m7	1 9 b3 11 5 13 b7
Phrygian	m7b9	1 b9 b3 11 5 b13 b7
Lydian	maj7#11	1 9 3 #11 5 13 7
Mixolydian	7	1 9 3 11 5 13 b7
Aeolian	m7	1 9 b3 11 5 b13 b7
Locrian	m7b5	1 b9 b3 11 #11 b13 b7
Harmonic Minor	mMaj7	1 9 b3 11 5 b13 7
BANK B — COLOUR	CHORD	DEGREES
Lydian Dominant	7#11	1 9 3 #11 5 13 b7
Altered	7alt	1 b9 b3 3 #11 b13 b7
Phrygian Dominant	7b9b13	1 b9 3 11 5 b13 b7
Melodic Minor	mMaj7	1 9 b3 11 5 13 7
Whole Tone	7#5	1 9 3 #11 b13 b7 6 notes
Diminished H-W	7b9	1 b9 b3 3 #11 5 13 b7 8
Minor Pentatonic	m7	1 b3 11 5 b7 5 notes
Bebop Dominant	7	1 9 3 11 5 13 b7 7 8

Modes with fewer than eight degrees wrap: the remaining columns continue into the next octave. Modes with eight fill the row exactly.



Temperament

A drone is the one context where tuning is fully exposed. Nothing moves, so beating between intervals is all you hear — which makes just intonation audibly, obviously different rather than a theoretical nicety.

Each pad on the grid shows how far its pitch sits from 12-TET, in cents. Changing temperament under a held drone glides.

SYSTEM	3RD / 5TH	
Just — 5-limit	+16 / +2	Small whole-number ratios: 3:2, 5:4, 6:5. Thirds and fifths lock and stop beating entirely. The default, and the reason to own this app.
Just — 7-limit	+16 / +2	Adds the septimal 7:4 and 7:6. Blue, slightly flat sevenths that sit beautifully under a horn.
Pythagorean	+8 / +2	Stacked pure fifths. Wide, singing thirds with a bright medieval edge.
1/4-comma Meantone	+14 / -3	Pure 5:4 thirds bought with narrowed fifths. Sweet and old.
12-TET	0 / 0	The neutral reference. Everything beats a little, nothing is out of place.
19-TET	-5 / -7	63.2¢ steps. Flat fifths, gorgeous thirds, a distinct minor/augmented split.
31-TET	+3 / -3	38.7¢ steps — near-perfect meantone with usable septimal intervals.
Harmonic Series	+14 / +2	Each degree snaps to the nearest partial 16–32 of the root. Overtone-locked, and strange in the upper degrees.



Timbre

Eight spectrum recipes, not eight presets of the same knobs. Each has its own partial series, inharmonic stretch, beating bias and filter scaling.

TIMBRE	PARTIALS	
Harmonium	10	Indian reed organ. Warm odd-leaning spectrum, air behind it, a slow shimmer between the reeds. Speaks quickly.
Pipe Organ	10	A drawn registration — 16' 8' 4' 2 $\frac{2}{3}$ ' 2' 1 $\frac{3}{5}$ '. Stable, wide, church-cold at the top. The least movement of the eight.
Shruti Box	10	Bellows and free reeds. Dark, thick, and it beats hard — the sound of a room breathing.
Bowed Strings	12	A full 1/n series under a soft filter. Rosin and horsehair. Slowest swell.
Bagpipe Drone	11	Cylindrical bore: odd harmonics dominate, evens barely there. Reedy, nasal, relentless.
Analog Bloom	11	Three-oscillator saw stack under a lazy filter. The 1970s idea of infinity.
Glass Choir	10	Stretched, hollow, mostly even partials. Sounds like light through a window.
Tanpura	14	Fourteen partials and a jawari bridge. Buzz is not a defect here, it is the instrument — this one has sitar in it before you arm anything.

Timbres carry a built-in sitar floor. Tanpura 45%, Bagpipe 14%, Shruti Box 10%, Harmonium 5%, the rest none. That floor adds to whatever you arm per tone, so the Sitar Depth slider does nothing at all on Pipe Organ until you arm a tone, and saturates almost immediately on Tanpura.



Tones

Eight degrees across four octaves — the same 32 pads as the top half of the Launchpad, in the same arrangement.

Anatomy of a pad

♭3	Scale degree, relative to the key.
F3	Note name and octave.
+16¢	Distance from where 12-TET would put it. Shows — when it lands within half a cent.
Bar	That tone's level.
Amber outline	The root of the key.
Violet dot + ring	Jawari armed. A sounding tone with jawari also goes pale rather than saturated.

Gestures

Click	Swell the tone in at the <i>New Tone Level</i> setting. Click again to let it go — it's a latch, so the second click can come whenever.
Drag ↑	Ride that tone's level live. Drag to the bottom and release to remove it.
↶-click	Arm or disarm jawari on that tone. Doesn't sound the tone.
Hover	Tooltip with the exact frequency in hertz.

Voicings

Fifteen of them, on the next page. Each fades whatever is sounding and builds a new stack, because a good drone is a specific voicing and not just "the notes of the chord".

TRANSPORT

Let go ⌘.	Everything fades at the <i>Fade</i> setting. The reverb tail continues.
Cut short ⇧⌘.	Everything gone in 1.2 seconds. Tail continues.
Silence ⌘⌘	Immediate. Voices killed, levels zeroed, and the reverb tail cleared.
Menu	The Drone menu holds all fifteen voicings, ⌘T next timbre and ⌘Y next temperament. The Pulse menu holds the clock and its presets.

STATUS BAR

Names whatever you last touched — the note and its exact frequency, the temperament's character, the voicing's intent — then falls back to the current key and reference pitch after a few seconds. On the right, a count of how many tones are holding.



Voicings

Fifteen starting points. Each fades whatever is sounding and builds a new stack in the mode you are already in.

Open Fifths	Root and fifth in three registers. Nothing to argue with — the safest bed under any soloist.
Full Chord	Every chord tone, one per register, spread wide. The most harmonically explicit.
Modal Spread	Chord tones low, colour tones high and quiet. The most to blow over.
Tanpura	The classic 5–1–1–1 cycle across all four registers. Weighted heavily to the bottom root.
Pedal Root	One low root and its octave. The strictest tonal centre.

BORROWED FROM TEN TRADITIONS

What is worth taking from people who have been holding drones for centuries is not the scale — Thrum already has the modes — but **which interval a tradition leaves out, and which register it puts the rest in.**

VOICING	FROM	
Highland	Scotland	Great Highland pipes: a bass drone an octave under two tenors, and no fifth anywhere.
Uilleann	Ireland	Three drones an octave apart, with the regulators' triad sitting on top.
Gaida	Bulgaria	The tonic under a second, in the same register — the rub village singing is built on.
Aitake	Japan	The shō's gagaku cluster: fourths and seconds over two octaves, every pipe the same weight, nothing in the bass.
Launeddas	Sardinia	The triple pipe — the tumbu's low drone with a fourth and a fifth laid over it.
Georgian	Georgia	Village polyphony over the bass: fifth, fourth and a bare seventh, and no third at all.
Didgeridoo	Australia	One fundamental and its own harmonic series, at harmonic-series levels. Almost all bottom.
Hardanger	Norway	The hardingfele's sympathetic understrings, ringing quietly in one high register.
Gyütö	Tibet	Gyütö chant, with the fifth down in the lowest register , where nobody else puts one.
Guqin	China	The qin's open strings — pentatonic, so no fourth and no seventh, over all four octaves.

Two of these want other settings moved with them. Gaida is a beating interval by design, so turn *Beating* down or the second and the tonic fight twice over. **Aitake** and **Hardanger** deliberately hold nothing in the bottom octave — reach for *Warmth* if the room needs a floor.

Each is described in scale degrees, and not every mode has the flavour of fourth or fifth the original was built on — so every degree falls back, and failing that takes whichever degree simply sits in that position. Two can still land on one pad in a mode that hasn't got both: whole tone has no perfect fourth or fifth, only the tritone between them.



Pulse

A tempo, and four arpeggiators running over the drone at the same time.

An arpeggio in Thrum is not a sequence of notes. It is an **accent on a voice that is already sounding**: the drone keeps holding exactly as it was, and the pulse rides a fast attack-and-decay envelope on top of whichever voice it is pointing at. Tap in a chord, press **Run**, and the chord starts to shimmer through itself. Stop the pulse and the drone is still standing, untouched.

Because the accent is *added* rather than substituted, a tone that is *not* being held will sound for as long as its ring — so a lane can play degrees that aren't in the drone at all, over the top of the ones that are.

TRANSPORT

TAP ⌘K	Tap four times in time with the room. The last tap is the downbeat, and it drops every lane back onto beat zero — so re-aligning to a drummer is one gesture.
Tempo	20–200 bpm. The – and + step it by one; there is a slider under <i>Pulse</i> in the parameter column too.
Beat dots	Four dots, one bar. The first is amber so you can find the downbeat without counting.
Presets	Eight whole-rig starting points, plus <i>all lanes off</i> and <i>realign lanes</i> .
Run ⌘P	Starts and stops the clock, and realigns every lane when it starts. PANIC stops the clock as well as the voices.

A LANE

Four identical ones. The number at the left arms it; the dot beside the number lights as that lane strikes.

Source	Held — the degrees you are holding, echoed into this lane's registers, so one chord tapped in low can arpeggiate in four octaves without tapping it in four times. Chord — the chord tones of the mode. Scale — every degree. Root — a pedal. Colour — the degrees the chord leaves out, which is where the good lanes usually are.
Pattern	Up · Down · Up·Down · Down·Up · Converge (outside in) · Diverge (inside out) · Pinwheel (a stride coprime with the set, so it covers everything in an order you can't predict) · Scatter (random, but the same random every time) · Chord (all at once — a pulse rather than an arpeggio).
Rate	$\times\frac{1}{2}$ $\times\frac{3}{4}$ $\times 1$ $\times 1\frac{1}{2}$ $\times 2$ $\times 3$ $\times 4$ $\times 6$ steps per beat.
Register	Which of the four octaves the lane works in — one, a pair, or all four.
Offset	Shoves the lane a quarter beat later. The cheapest way to turn two lanes that are locked together into two that are chasing each other.
Level	How far above the drone that lane sits.



Rates, and the eight presets

Four lanes at four rates is where the hypnosis is. Getting there is mostly a matter of not letting two of them agree.

RATES ARE RATIOS, NOT NOTE VALUES

The interesting divisions here are the ones that fight. A lane at $\times 1$ against a lane at $\times 1\frac{1}{2}$ takes six beats to come back around; $\times 1$ against $\times 6$ with a five-note set and a seven-step accent takes long enough that you stop hearing a loop at all. That drift is the whole point, and it is why the rates are written as ratios rather than as eighths and sixteenths.

Give each lane its own **register** and they never collide — four lanes, four octaves, four speeds. Give two lanes the same register and the same rate but different **offsets** and they chase each other instead.

Arpeggiate a tone with jawari armed. A plucked note through the bridge buzz is what a sitar actually is, and it is the one combination in Thrum where two features add up to a third thing.

PRESETS

PRESET	BPM	
Pulse	64	One lane, one note a beat. The plainest thing that still moves.
Three : Two	68	Two lanes, one and a half against one. Comes round every six beats.
Tanpura	52	A pedal underneath, the held chord walking above it.
Gamelan	76	Four registers, four rates, chord tones low and colour up top.
Phase	96	Same figure, four rates, four starting points. Give it two minutes.
Rain	60	A slow root, and scatter falling through the top two octaves.
Undertow	48	Half-time and three-quarter time, low and slow. Barely an arpeggio.
Swarm	108	Everything at once, quietly. Individual notes stop being audible.

The clock runs off the main thread, so the pulse doesn't stutter when the window redraws. Step numbers are absolute positions on that clock rather than a running index, which means a lane keeps its place in the bar even if you add a note to the chord underneath it mid-figure.



Parameters

Twenty-eight sliders in five groups. Hovering any of them shows what it does — the label fades in without moving anything — and the value on the right is live.

VOICE — HOW A TONE BEHAVES

CONTROL	RANGE	DEFAULT	
Swell	0.4–40 s	7 s	How long a tone takes to bloom to full after you tap it in. Scaled per timbre — reeds speak faster than bowed strings.
Fade	0.4–60 s	11 s	How long a tone takes to disappear when you let it go.
Beating	0–100%	50%	Detune between each partial's oscillator pair, up to about 1.1 Hz. This is the breathing. At zero the drone goes glassy and still.
Drift	0–100%	45%	Slow independent pitch wander per voice, up to ± 9 cents, on prime-numbered cycles.
Motion	0–100%	55%	Depth of the slow filter, tremolo and panning movement. Nothing should ever be perfectly still.
Sitar Depth	0–100%	55%	The jawari amount written to a tone when you arm it — <i>not</i> a global effect. Also updates tones already armed. Explained in full under <i>Jawari</i> .
New Tone Level	0–100%	72%	Loudness a tone swells to when you tap it in.

TONE — THE SPECTRUM

CONTROL	RANGE	DEFAULT	
Brightness	0–100%	52%	Per-voice lowpass, 320 Hz to 9 kHz, before the slow filter sweep.
Warmth	0–100%	55%	Low shelf at 165 Hz, up to +8 dB. The weight under everything.
Presence Cut	0–100%	60%	Dip at 3.2 kHz, up to –7 dB. The most important control here. Energy around 2–5 kHz is what makes a long drone tiring; this is the guard against it.
Air	0–100%	40%	High shelf at 9 kHz, –3 to +5 dB. Soft top, not edge.
Saturation	0–100%	28%	Gentle tanh drive ahead of the reverb. Adds the overtones a pure sum of sines can't produce.



Parameters, continued

SPACE — THE ROOM

CONTROL	RANGE	DEFAULT	
Decay	1–32 s	14 s	RT60 of the feedback delay network. Ten to thirty seconds blurs individual events into a continuous field.
Wet	0–100%	42%	How much of the tail you hear.
Damping	0–100%	45%	High-frequency absorption in the tail. Up is darker and further away.
Size	0.6–1.7	1.15	Scales the delay-line lengths. Bigger is slower and more diffuse.
Width	0–2	1.25	Mid/side spread. The fundamental stays centred; harmonics open outward. 1 is untouched, 0 is mono.
Field Drift	0–100%	50%	A 97-second rotation of the reverb field left to right. You only notice it if you wait.
Field Radius	0.4–6 m	1.6 m	<i>Spatial mode only.</i> How far out the ring of tones sits. Under a metre is a helmet; past three it is a room you are standing in.
Field Lift	0–45°	22°	<i>Spatial mode only.</i> How far apart the low and high octave rings are pushed vertically. At zero everything sits on one ring at ear level.

MASTER

CONTROL	RANGE	DEFAULT	
Swell Ride	0–100%	100%	Rides everything sounding at once. Live dynamics for the whole drone — the thing to grab when the soloist drops to a whisper.
Output	0–100%	85%	Master level, after the limiter.



Parameters · the pulse

The six that shape the arpeggios. *Strike* and *Ring* are the two that change the character most — they are the difference between a marimba inside the drone and the chord simply breathing.

PULSE — THE ARPEGGIOS

CONTROL	RANGE	DEFAULT	
Tempo	20–200 bpm	68	The beat every lane divides. Tap it in rather than typing it.
Strike	2–600 ms	35 ms	How sharply an arpeggiated note arrives. Short is a pluck, long is a bow. Under about 10 ms a stack of pure partials starts to click.
Ring	0.05–9 s	1.4 s	How long it takes to fall back into the drone. Past a couple of seconds the notes stop being separate and the chord simply breathes.
Accent	0–100%	70%	How far above the drone the arpeggio sits. Low is a shimmer inside the chord; high is a part.
Swing	0–70%	0%	Pushes every other step late by a fraction of <i>that lane's own</i> step, so lanes at different rates all swing together.
Unsteady	0–100%	22%	Velocity wobble, the same every time round. Without a little of it a long cycle reads as a machine.

LAUNCH CONTROL PANEL

Click a name	Starts MIDI learn for that parameter — move a knob to bind it. The binding persists across launches.
Cancel learn	Abandons a pending learn.
Reset map	Back to the factory layout for whichever model is attached.



Flow • the button

Press it and Thrum takes itself somewhere. It is meant to be worked through — put it on and go read something.

Modes turn over, voicings rebuild, timbres change, arpeggios arrive and leave, jawari appears on a tone and later disappears, and two dozen controls are sliding at any given moment.  does the same as the button. It is a **different journey every time** — the generator is reseeded on every start, not just when the app launches.

WHY IT DOES NOT JAR

Nothing is set, only ramped. Every change is a slide of twenty to ninety seconds on a curve with zero velocity *and* zero acceleration at both ends, so it has no edge to notice at either side. The shortest slide Flow will open is nine seconds; quicker than that reads as an event rather than a drift.

The jumps are hidden under a breath. Swapping a timbre recomputes every partial, and moving the key sweeps every sounding pitch — at full level, both lurch. So Flow leans on *Swell Ride* first, makes the change with the drone down at a fifth of its level, and brings it back over fourteen seconds. Key moves also glide over seven and a half seconds rather than the half-second a change you made yourself gets, which turns a fifth into a modulation.

NOTHING LINES UP

GESTURE	EVERY	
Controls drift	10–20 s	Two to four sliders start new slides, so several always overlap and none is the event.
Arpeggios shift	1–3 min	A lane changes rate, pattern, source or register — or the pulse arrives out of silence, or thins and leaves.
Voicing rebuilds	2–4½ min	Already a crossfade: the old stack fades at <i>Fade</i> as the new one swells at <i>Swell</i> .
Jawari moves	2–5 min	A sitar buzz armed on a sounding tone, or taken off one.
Mode turns over	2½–5½ min	Restful modes only — Locrian, Altered, Diminished and Whole Tone are good to play <i>over</i> and wearing to sit <i>inside</i> .
Timbre changes	3½–8 min	Under a breath.
Key moves	5–10 min	By a fourth or a fifth — the moves sharing most notes with where you were. Sometimes a register instead.
Temperament	7–15 min	Rare, and it glides.

Each runs on its own clock at its own tempo, so the arrangement never falls into a pattern — the engine's prime-numbered modulation, one level up.

Two things Flow will not do. It never touches **Output** — the volume is yours. And **Presence Cut** has a floor, because energy around 3 kHz is what makes a long drone tiring and this is the one mode that runs for hours unattended. Everything else roams inside a range chosen to stay pleasant for an hour rather than to show what is possible.

Turning Flow off leaves everything where it drifted to — no snap back, because where it got to is usually better than where you started. Anything you touch while it runs, it will drift away from again in its own time.



Spatial

The drone placed around your head, and kept there when you turn it. AirPods Pro or AirPods Max.

Instead of one stereo mix, spatial mode renders **sixteen mono buses** positioned around you and rendered binaurally. The grid already has two axes worth putting in a room, so where a tone sits is a fact about the music rather than a decoration: the **eight columns are compass points** with the tonic dead ahead, and the **four rows are two rings** — the low two octaves slung below the ear line, the high two lifted above it.

Which means an arpeggio walking up a column *climbs* as well as rising in pitch, and a lane walking across the mode *orbits* you. Four lanes at four rates become four orbits at four speeds, which is the thing the pulse was built for.

THE CONTROLS

Spatial ⌘S	In or out. Everything else in the app behaves identically either way.
Head ⌘H	Reads your AirPods' orientation and counter-rotates the listener, so the drone stays where it is in the room as you turn. Asks for Motion permission the first time; declining leaves the field head-locked and nothing else changes.
Recenter ⌘R	Takes wherever you are looking now as straight ahead. AirPods yaw drifts, so this is the button you will actually use — and the one to hit if the field ever feels tilted.
Radar	Top-down view. Two rings of eight, inner is the low octaves; each dot lights with what that bus is putting out. The line from the centre is where your head is pointing, and the tick at the top is straight ahead.
Field Radius · Lift	Under <i>Space</i> in the parameter column. How far out the rings sit, and how far apart the two octave tiers are pushed vertically.

Two things to get right or it will not sound like much. Set **macOS's own Spatial Audio to Off** for your AirPods — two lots of HRTF smear each other. And **turn Wet down**: the reverb tail deliberately bypasses the spatial stage, because a thirty-second diffuse tail has no location to be placed at, so it stays a stereo bed and does not rotate with your head. The more of it you have, the less the field localizes.

WHAT IT COSTS

Almost nothing. The sixteen HRTF instances add about **0.2% of one core** on top of the engine, measured with the app's own graph. Spatial mode is not the thing to switch off if you are chasing a dropout.



The sitar control, in detail

A sitar's bridge is flat and wide, so the string grazes it as it vibrates and sheds a cascade of upper harmonics locked to the fundamental. That grazing is called *jawari*, and it is the reason a sitar shimmers.

IT IS PER-TONE, AND IT IS ARMED RATHER THAN DIALLED

Sitar Depth is the value written to a tone when you arm it. With nothing armed, moving the slider does nothing audible. Moving it while tones are armed updates all of them together.

In the app	⌘-click a pad.
On the Launchpad	Hold SITAR and tap tones. Pressing into the SITAR pad sets the depth live, from 15% to 100%.
By timbre	Tanpura, Bagpipe, Shruti Box and Harmonium carry a floor that applies to every voice with nothing armed.

WHAT THE DEPTH ACTUALLY DRIVES

Three things move together as you raise it:

AT 0 → 100%

Phaser mix 0 → 75%	Four cascaded allpass stages whose coefficient is swept by an LFO on a prime-derived period, unique per voice — so no two sitar tones ever swirl together. This is the "slight phase" part.
Comb feedback 0.55 → 0.83	A comb exactly one period of the fundamental long, so it reinforces the harmonic series rather than fighting it. This is the sympathetic ring.
Buzz mix 0 → 30%	High-passed at 700 Hz, so only the bright edge is added and none of the mud.

Changes glide in over a second or two rather than switching.

Use it on one voice, not all of them. The effect is per-tone by design. Arm a single high tone and leave the rest clean and you get the tanpura's shimmering top string over a steady bed. Arm everything and it smears.

TWO HONEST LIMITATIONS

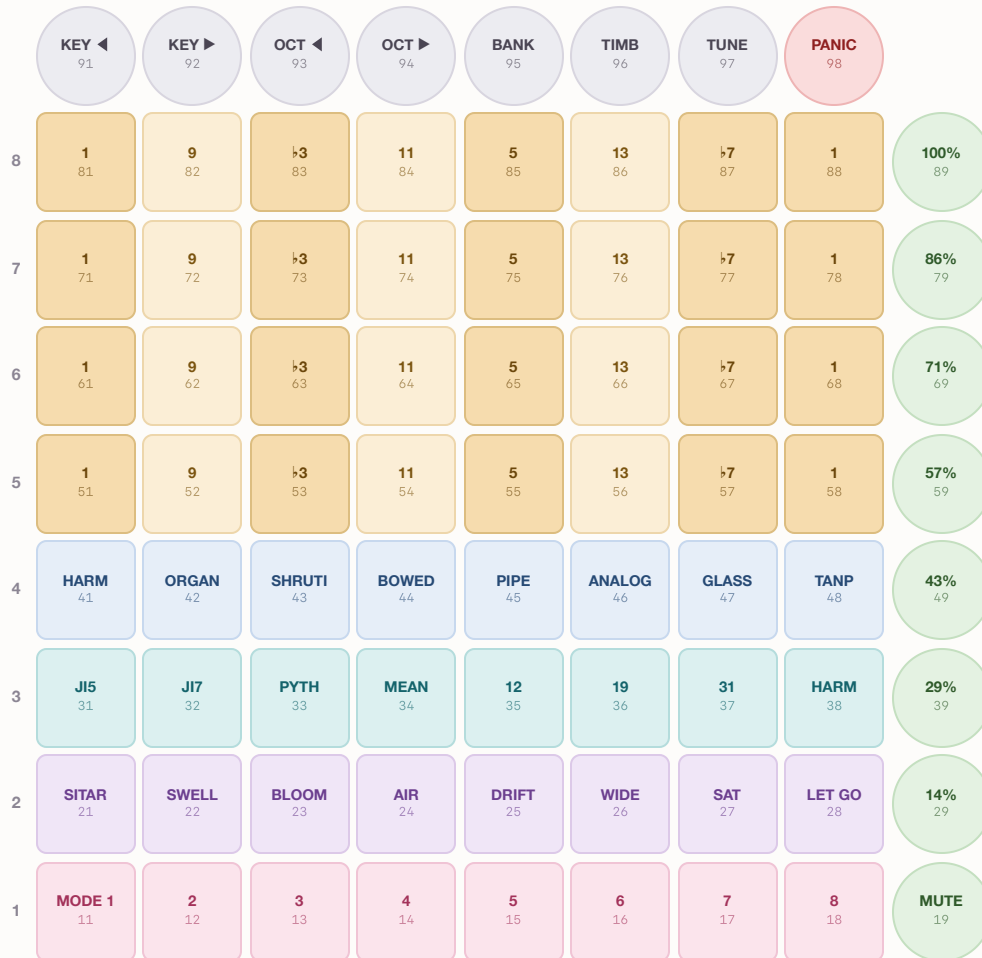
The waveshaper is an *odd* function, so it produces odd harmonics only — 3rd, 5th, 7th. A real jawari gives you the whole series, so this reads a little hollow beside the real thing.

There is a hard clip inside the comb's feedback loop, needed because the shaper grows with level rather than compressing. Hard clipping produces harmonics above Nyquist that fold back as aliasing, so at high depth on the upper rows you may hear a gritty metallic edge that isn't musical buzz. If it bothers you, keep the depth moderate on the top octave.



Every key

Plug it in and Thrum takes over — programmer mode is set automatically, no template to load. All 81 keys do something.



Chord tone Scale tone Timbre Temperament Modifier (hold) Mode Output ladder Global

Small numbers are programmer-mode note or CC values. Degree labels on the tone rows show D Dorian; they change with the mode. The logo LED (CC 99) is an output indicator — it breathes with the level and takes no input.

The grid tells you what it's doing. Tone LEDs track the actual envelopes, so a pad brightens as its tone swells in over seven seconds rather than snapping on when you tap it. A tone with jawari armed goes pale instead of saturated.



Rows 8–5 · the tones

Thirty-two pads: eight scale degrees across, four octaves up. Row 5 is the lowest, row 8 the highest. Chord tones glow amber when idle, the rest of the scale sits nearly dark.

Tap	Swell the tone in. Tap again to let it go — a latch, so the second tap can come whenever.
Press into it	Aftertouch rides that tone's level, live.
Hold SITAR + tap	Arm or disarm jawari on that tone.
Hold LET GO + tap	Drop that tone outright.

A "tap" is a release inside 0.45 s with no real pressure behind it. Anything heavier or slower is read as a level ride and leaves the tone where your finger left it — which is why **hold LET GO + tap** exists as an unambiguous kill.

Row 2 · the modifiers

All eight are hold-and-press. They push a parameter while held and put it back on release, so nothing is left somewhere you didn't intend.

PAD	HELD
SITAR	Tone pads become jawari toggles. Pressure sets the depth, 15% to 100%.
SWELL	Rides the whole drone up and down, 12% to 100%. Springs back to full on release. The main expressive control.
BLOOM	Opens the reverb — wet, size and decay together, up to a 30-second tail.
AIR	Brightens: voice filter and the high shelf together.
DRIFT	More wander — drift, motion and beating together.
WIDE	Opens the stereo field: width to maximum and the reverb rotation to full.
SAT	More saturation, up to 95%.
LET GO	Tap for a graceful fade of everything · press past 70% to cut it short in 1.2 s · hold and tap tones to drop just those.



Rows 4, 3 and 1 · selection

Row 4	The eight timbres, left to right, in the same order as the app. Lit in the timbre's own colour, current one at full brightness.
Row 3	The eight temperaments, in the same order as the app. Cyan.
Row 1	Eight modes — whichever bank is active. Lit in each mode's colour. Sounding tones glide to the new mode rather than jumping.

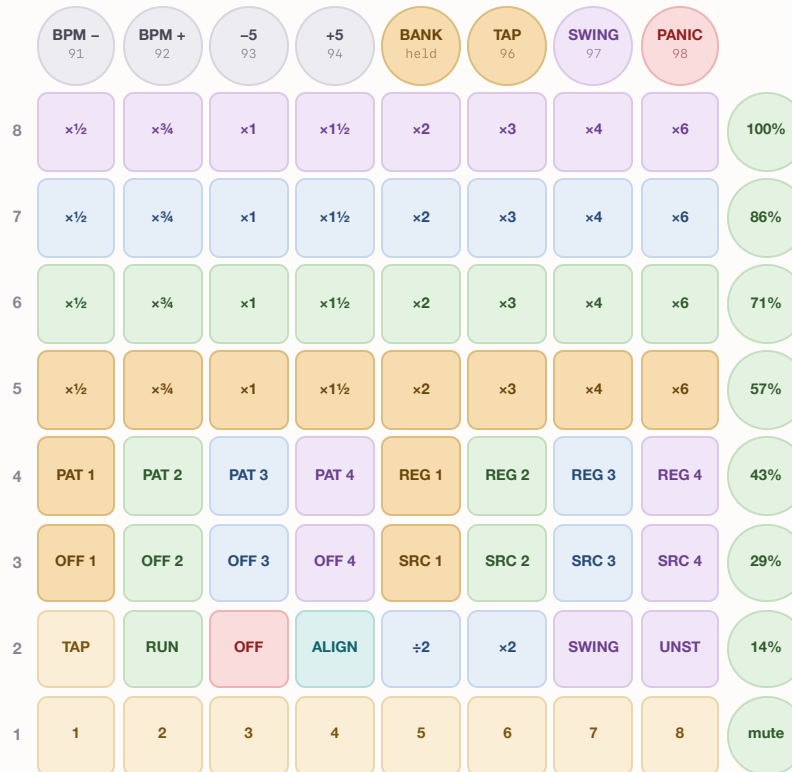
Top row and right column

KEY ◀ ▶	Move the key by a semitone. Held tones glide.
OCT ◀ ▶	Move the whole grid by an octave, 1–6.
BANK	Tap to swap row 1 between mode bank A (diatonic) and B (colour) — dim blue for A, violet for B. Hold it and the whole board becomes the arpeggiator page, on the next page. It only cycles the bank if you let go without having pressed anything else.
TIMB	Next timbre. Lit in the current timbre's colour.
TUNE	Next temperament.
PANIC	Immediate silence, reverb tail included.
Right column	An eight-step output ladder, lit from the bottom up — green at the bottom, amber at the top, readable in the dark. The bottom pad is a mute: every tone keeps holding behind it.
Logo	Output indicator only. Breathes with the level, in the current mode's colour.



Hold BANK · the arpeggiator

A second layer over the whole board. Nothing was taken away to make room for it: tapping BANK still cycles the mode bank, and it only does so if you release it without having pressed anything else.



Rows 8–5 one lane each — 5 is lane 1, 8 is lane 4 Row 4 pattern, then register Row 3 offset, then source Row 2 transport

Row 1 the eight presets

HOW IT PLAYS

Rows 8–5	One row per lane, one pad per rate. Pressing a rate arms the lane as well as setting it; pressing the rate already lit turns the lane off. Keep a finger here — retuning the polyrhythm live is the most useful gesture on the page.
Rows 4 · 3	Lane <i>N</i> is always column <i>N</i> , so the lanes read as four vertical stripes in the app's own four colours. Each pad cycles that lane's pattern, register, offset or source forward by one.
Row 2	TAP · RUN · all lanes off · realign · halve · double · swing · unsteady. RUN breathes on the beat while the clock is going.
Row 1	The eight presets. One press replaces all four lanes and the tempo.
Top row	Tempo by ±1 and ±5, and TAP sitting right beside BANK so one hand holds while the other taps. PANIC stays PANIC — the one key that means the same thing on every layer.



Launch Control

Plug one in and its knobs land on Thrum's parameters in the order the app shows them. Nothing to configure.

Both models	The Launch Control (16 knobs) and the Launch Control XL (24 knobs + 8 faders) are told apart automatically and get different factory-template CC maps.
Soft takeover	A knob does nothing until it passes through the parameter's current value, so grabbing one never jumps the sound mid-performance.
MIDI learn	Click a parameter's name in the app, then move a knob. The binding is saved and survives relaunching.
Pads	Trigger the first six voicings, plus let-go and panic. Capped at six on purpose: there are fifteen voicings now, and letting them fill all eight pads would cost the panic button.

Both devices report themselves as "Launch Control"; only the XL appends "XL", and that is what Thrum keys off. If your device is on a non-factory template, learn the knobs once and it will remember.

DEFAULT KNOB ORDER

Swell · Fade · Beating · Drift · Motion · Sitar Depth · New Tone Level · Swell Ride
Brightness · Warmth · Presence Cut · Air · Saturation · Width · Field Drift · Output
Decay · Wet · Damping · Size
Tempo · Strike · Ring · Accent · Swing · Unsteady

A 16-knob Launch Control reaches the first sixteen; the XL's 32 controls reach all twenty-six. The pulse controls sit last so that adding them didn't move anybody's existing knobs.



If something sounds wrong

Crackle, ticks, dropouts	Check the DSP readout in the top bar. If it shows drops in red, the engine is missing its deadline. Nothing sounding? Then it isn't Thrum. <i>Never run a Debug build</i> — the engine is roughly 40× slower without optimisation and cannot keep up.
Sound continues after silencing	That's the reverb tail, which can run thirty seconds. PANIC clears it; Let go and Cut short deliberately don't.
Grid is dark or unresponsive	Check the device pill in the top bar. Unplug and replug — Thrum rescans on any MIDI change and re-enters programmer mode.
Launchpad in the wrong mode	Thrum uses the "LPX MIDI" port, not "LPX DAW". If another app has claimed the device or left it in a different mode, quit that app and replug.
Sitar Depth does nothing	Expected on timbres with no floor. Arm a tone first: <code>⌘</code> -click, or hold SITAR and tap.
A tone won't tap off	You're pressing too hard or too long and it reads as a level ride. Hold LET GO and tap it instead.
Pulse is running but silent	Every lane defaults to the Held source, so it needs tones held to arpeggiate. Either tap a chord in, or switch a lane to Chord or Scale , which draw from the mode whether or not anything is sounding.
Arpeggio sounds mechanical	Raise Unsteady , and give two lanes rates that don't divide into each other — $\times 1$ with $\times 1\frac{1}{2}$ rather than $\times 1$ with $\times 2$. Equal rates lock; unequal ones drift.
Arpeggio buries the drone	Lower Accent , or lengthen Ring so the notes overlap into the chord instead of standing out of it.
Spatial sounds like ordinary stereo	Almost always macOS's own Spatial Audio also being on for the AirPods — set it to Off. Failing that, Wet is high enough that the un-spatialized reverb bed is drowning the field; bring it down.
The field turns the wrong way	Hit Recenter facing forward. If a tone still <i>follows</i> your head instead of staying put, that is a genuine bug rather than a setting — the sign convention is measured in Tools/axis.
Head tracking never starts	Only AirPods Pro and AirPods Max report orientation. If it was denied, Thrum appears under Privacy & Security → Motion & Fitness.
Flow changed something I liked	Turn it off — the  or  — and everything stays exactly where it had drifted to. Nothing snaps back.
Everything sounds still	Beating, Drift or Motion is at zero. Those three are what keep it alive.
Tiring after a while	Raise Presence Cut . Energy around 2–5 kHz is what wears the ear out over twenty minutes.

A drone wants patience. Most of what Thrum does happens on cycles of forty seconds to a minute and a half. If you audition it for ten seconds at a time you will hear a pad; give it five minutes and you will hear the instrument.